BUSINESS INTELLIGENCE LAB

UNIT 2 | SESSION 7

EXPERIMENT 7

Integration of Advanced Analytics with BI Reports using Microsoft Power BI

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1. AIM

To integrate advanced analytics with Business Intelligence reports using Microsoft Power BI and generate business insights, strategic recommendations, and executive presentations that support organizational decision-making.

2. OBJECTIVES

- To reuse the existing Power BI reports, Executive Dashboard, and Business Scorecard developed in Experiments 5 and 6.
- To analyze sales performance and profitability across regions.
- To analyze profitability by product category and customer segment.
- To analyze monthly and yearly sales trends and compare sales growth with profit growth.
- To identify business strengths, weaknesses, and opportunities supported by Power BI evidence.
- To prepare a Business Insight Report with findings, recommendations, and supporting screenshots.
- To prepare a Recommendation Summary Sheet and a 6-slide Executive Presentation outline.
- To justify every recommendation with data-driven evidence and measurable KPIs.

3. TOOLS / SOFTWARE REQUIRED

1. Microsoft Power BI Desktop.
2. Existing Power BI project file (.pbix) from Experiments 5 and 6, saved as BI_Experiment_7.pbix.
3. Global Superstore Dataset.xlsx.
4. Power Query Editor for data inspection, if required.
5. DAX (Data Analysis Expressions) for KPI and analytical measures.
6. Microsoft PowerPoint for the Executive Presentation.
7. Microsoft Word / Excel for the Business Insight Report and Recommendation Summary Sheet.

4. DATASET USED

Dataset: Global Superstore Dataset (Kaggle).

Source: https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset

The experiment reuses the same Power BI file and data model built in Experiments 5 and 6. No new project was created and no existing dashboards were rebuilt. The dataset covers sales transactions from 2014 to 2017 across global regions, product categories, and customer segments.

The important fields used in this experiment are:

- **Order ID:** Identifies individual orders.
- **Order Date:** Supports monthly and yearly trend analysis.
- **Sales:** Represents sales revenue.
- **Profit:** Represents profit generated from sales.
- **Region / Country:** Represent geographical sales areas.
- **Category:** Identifies product categories (Furniture, Office Supplies, Technology).
- **Segment:** Identifies customer segments (Consumer, Corporate, Home Office).
- **Customer Name:** Identifies customers.

5. THEORY

5.1 Advanced Analytics in Business Intelligence

Advanced Analytics in Business Intelligence focuses on transforming data into actionable insights. While traditional BI reports present information, advanced analytics helps identify trends, opportunities, business challenges, and improvement areas. Organizations use analytical reports and dashboards to support strategic planning and decision-making.

5.2 Analytical Techniques Used

Power BI enables advanced analytics through the following techniques, all of which are applied in this experiment:

- **KPI Analysis:** Monitoring Total Sales, Total Profit, Total Orders, and Profit Margin.
- **Trend Analysis:** Studying monthly and yearly sales movement over time.
- **Comparative Analysis:** Comparing sales and profit across regions, categories, and segments.
- **Ranking:** Ordering regions and customers by sales or profit to find leaders and laggards.
- **Segmentation:** Filtering performance by region, category, and year using slicers.
- **Target Analysis:** Comparing actual results with predefined targets in the Business Scorecard.

5.3 Business Scenario

Global Superstore management has observed that sales are increasing but profitability is not growing at the expected rate. Senior management requires an analytical review of business performance. Acting as Business Intelligence Consultants, the Power BI reports and dashboards are used to identify issues, discover opportunities, and recommend corrective actions.

5.4 Strengths, Weaknesses, and Opportunities

Term	Meaning in this analysis
Strength	An area where the data shows the business is performing well, such as a region with high sales and healthy profit.
Weakness	An area where the data reveals poor performance, such as negative profit or high sales with a very low margin.
Opportunity	An area where a specific management action could improve a measurable KPI.

5.5 Analytical Measures

The following measures from Experiment 6 are reused. Profit Margin is the key measure for testing whether profit is keeping pace with sales.

Profit Margin (%)

```
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
```

Year-over-Year Sales Growth (%)

```
YoY Growth % = (Sales Current Year - Sales Previous Year) / Sales
Previous Year
```

Share of Total

```
Sales Share % = Region Sales / Total Sales
Profit Share % = Region Profit / Total Profit
```

A region whose profit share is much lower than its sales share is earning less than the company average margin. This comparison is used to identify high-sales, low-profit areas.

6. PROCEDURE

Part A: Analyze Existing Reports and Dashboards

Step 1: Opened Microsoft Power BI Desktop, loaded the existing Global Superstore project from Experiments 5 and 6, and saved a separate copy as **BI_Experiment_7.pbix**. The dataset loaded correctly and all report pages were available: Geographical Sales Analysis Dashboard, Relational Report, Matrix Report, Comparative Analysis, Dashboard, Executive Dashboard, and Business Scorecard.

Before filtering, the Executive Dashboard and Matrix Report give the overall business position for 2014 to 2017:

KPI	Value	Source visual
Total Sales	9,596,822.81 (9.60M)	Matrix Report total, Scorecard card
Total Profit	1,101,816.89 (1.10M)	Matrix Report total, Scorecard card
Total Orders	About 25,750 (26K)	Scorecard card (5.15 × 5,000 target)
Profit Margin	11.5%	Total Profit / Total Sales

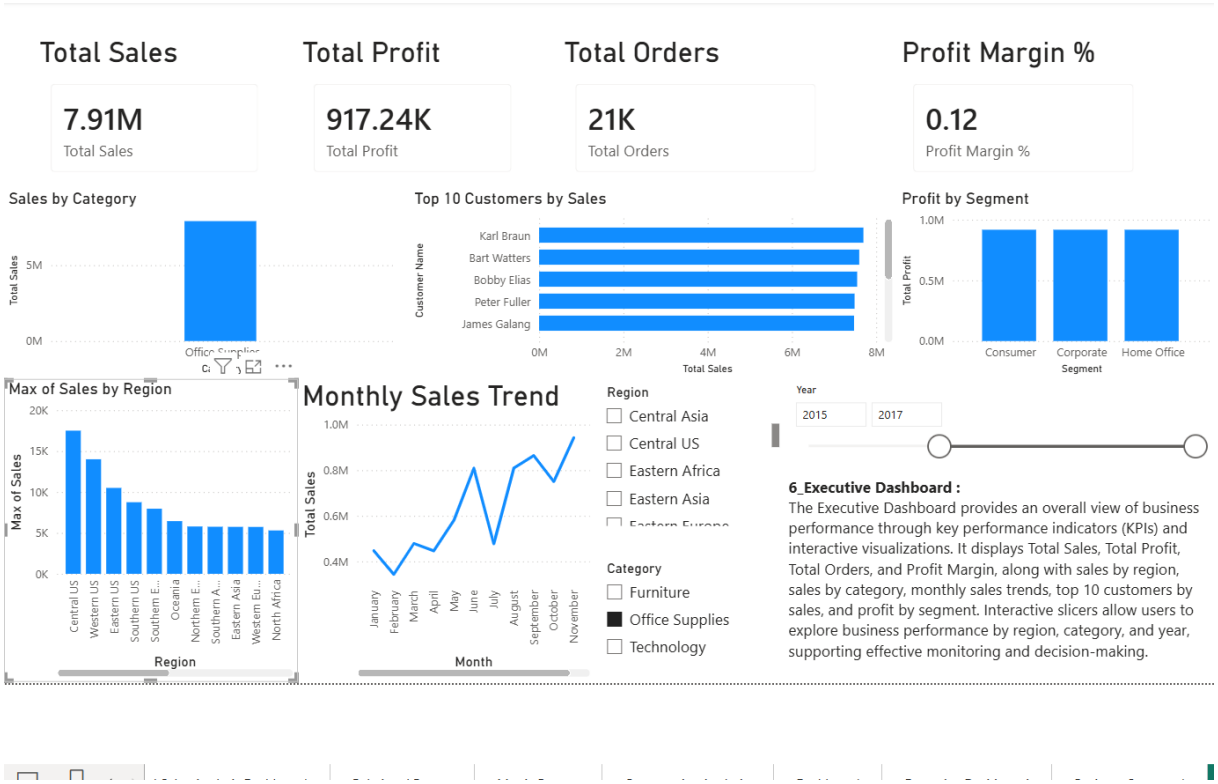


Figure 1: Executive Dashboard (Year slicer 2015 to 2017) showing Sales 7.91M, Profit 917.24K, Orders 21K and Profit Margin 0.12.

Step 2: Analyzed Sales Performance by Region using the Sum of Sales by Region chart, the Profit by Region chart, and the Matrix Report, which shows exact Sales and Profit per region.



Figure 2: Geographical Sales Analysis Dashboard showing Sum of Sales by Region.

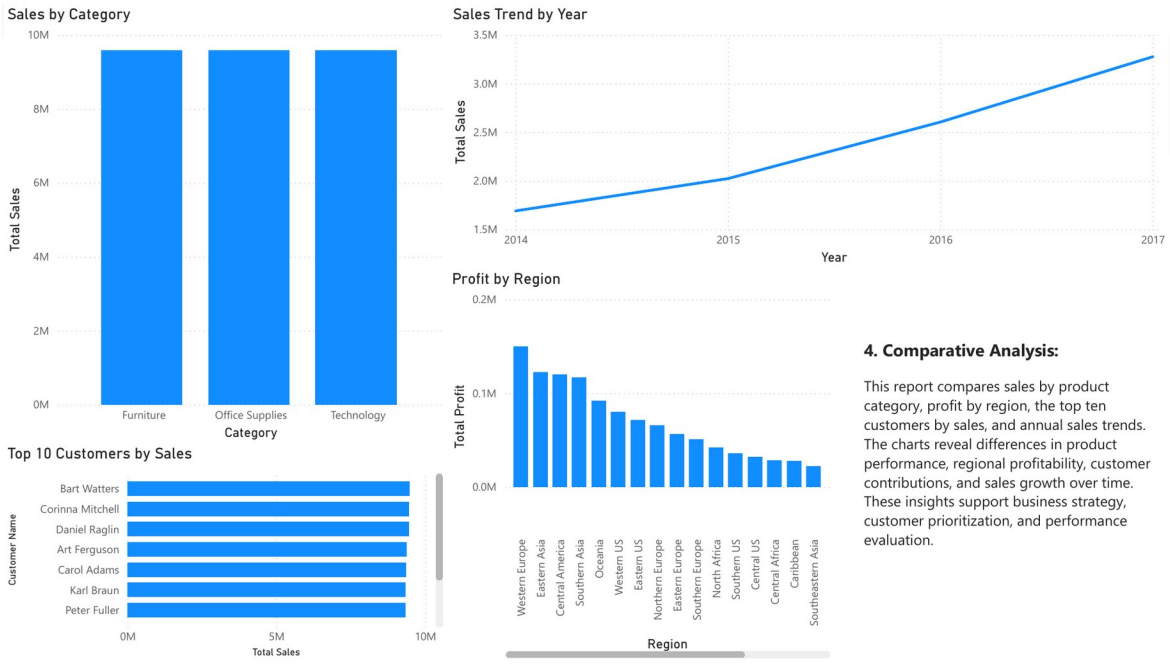


Figure 3: Comparative Analysis page showing Profit by Region and Sales Trend by Year.

The exact regional figures read from the Matrix Report are:

Region	Sales	Profit	Margin	Sales share	Profit share
Western Europe	1,294,590.88	150,495.79	11.6%	13.5%	13.7%
Central America	934,250.46	120,447.84	12.9%	9.7%	10.9%
Oceania	841,748.06	92,160.77	10.9%	8.8%	8.4%
Southeastern Asia	665,269.66	22,400.65	**3.4%**	6.9%	**2.0%**
Southern Asia	659,911.20	116,814.43	17.7%	6.9%	10.6%
Eastern Asia*	about 634K	122,821.14	19.4%	6.6%	11.1%
South America	459,210.18	17,962.50	**3.9%**	4.8%	**1.6%**
Southern Europe	451,117.70	51,121.97	11.3%	4.7%	4.6%
Western Asia	238,084.48	**-39,604.61**	**-16.6%**	2.5%	-3.6%
Caribbean	237,740.13	27,696.57	11.6%	2.5%	2.5%
Eastern Europe	233,265.06	56,523.63	24.2%	2.4%	5.1%
North Africa	170,983.92	42,366.81	24.8%	1.8%	3.8%
Western Africa	135,797.11	**-37,294.52**	**-27.5%**	1.4%	-3.4%
Central Africa	111,970.26	28,520.73	25.5%	1.2%	2.6%
Eastern Africa	100,631.33	18,191.54	18.1%	1.0%	1.7%
Southern Africa	79,666.26	17,456.04	21.9%	0.8%	1.6%
Central Asia*	small (under 50K)	negative	**-46.4%**	under 0.5%	negative

Note: Margin and share percentages were calculated from the Matrix Report values against the overall totals (Sales 9,596,822.81; Profit 1,101,816.89). Rows marked * come from the Business Insights page (Region table and Profit Margin % by Region chart), which was added in this experiment; Central Asia's exact sales and profit are not shown in the captured visuals, only its margin and its position on the scatter chart.

Country	Total Sales	Total Profit	Sum of Quantity
Afghanistan	**659,911.20**	**116,814.43**	**5084**
Herat	659,911.20	116,814.43	5084
Jalalabad	659,911.20	116,814.43	5084
Kabul	659,911.20	116,814.43	5084
Kandahar	659,911.20	116,814.43	5084
Albania	**451,117.70**	**51,121.97**	**4381**
Algeria	**170,983.92**	**42,366.81**	**1603**
Angola	**111,970.26**	**28,520.73**	**878**
Argentina	**459,210.18**	**17,962.50**	**6040**
Armenia	**238,084.48**	**-39,604.61**	**2999**
Australia	**841,748.06**	**92,160.77**	**7043**
Austria	**1,294,590.88**	**150,495.79**	**12282**
Azerbaijan	**238,084.48**	**-39,604.61**	**2999**
Bahrain	**238,084.48**	**-39,604.61**	**2999**
Bangladesh	**659,911.20**	**116,814.43**	**5084**
Barbados	**237,740.13**	**27,696.57**	**3539**
Belarus	**233,265.06**	**56,523.63**	**2145**
Belgium	**1,294,590.88**	**150,495.79**	**12282**
Belize	**934,250.46**	**120,447.84**	**11486**
Benin	**135,797.11**	**-37,294.52**	**1880**
Bhutan	**659,911.20**	**116,814.43**	**5084**
Bolivia	**459,210.18**	**17,962.50**	**6040**
Bosnia and Herzegovina	**451,117.70**	**51,121.97**	**4381**
Botswana	**79,666.26**	**17,456.04**	**609**
Brazil	**459,210.18**	**17,962.50**	**6040**
Bulgaria	**233,265.06**	**56,523.63**	**2145**
Burkina Faso	**135,797.11**	**-37,294.52**	**1880**
Burundi	**100,631.33**	**18,191.54**	**991**
Cambodia	**665,269.66**	**22,400.65**	**6207**
Cameroon	**111,970.26**	**28,520.73**	**878**
Total	**9,596,822.81**	**1,101,816.89**	**97675**

3. Matrix Report

This report summarizes Total Sales, Total Profit, and Quantity across Region, State, and City using the Geo_Hierarchy. Drill-down and drill-up operations allow users to explore geographical performance at different levels. It helps compare regions, identify high-performing cities, and support regional business planning.

Figure 4: Matrix Report showing Total Sales, Total Profit and Quantity by country (country rows carry their region totals).

Observations:

- **Highest sales:** Western Europe (1.29M), which is also the most profitable region (150.5K). Here the highest-sales region is also the highest-profit region.
- **Second tier:** Central America (934K sales, 120K profit) and Oceania (842K sales, 92K profit) follow, both with healthy margins near the company average.
- **High sales, weak profit:** Southeastern Asia ranks 4th in sales (665K) but earns only 22.4K profit (3.4% margin). South America sells 459K but earns only 18.0K (3.9%).
- **Loss-making:** Western Asia (-39.6K) and Western Africa (-37.3K) together lose about 76.9K, which wipes out about 7% of total company profit. Central Asia has the worst margin of all regions (-46.4%), although on very small sales.
- **Hidden strength:** Eastern Asia earns 122,821.14 profit on about 634K sales, a 19.4% margin. It is the second most profitable region despite ranking only 6th in sales.
- **Below average in the US:** Central US earns only an 8.1% margin, the lowest of the US regions.
- **Small but very profitable:** Central Africa (25.5%), North Africa (24.8%), Eastern Europe (24.2%) and Southern Africa (21.9%) earn margins about double the company average on low sales volumes.

Applying the Region slicer on the Executive Dashboard confirmed these patterns at the filtered level:

Region filter (Year 2015 to 2017 unless stated)	Sales	Profit	Orders	Margin
All regions (2015 to 2017)	7.91M	917.24K	21K	0.12
Caribbean (2014 to 2017)	237.74K	27.70K	856	0.12
Central US	315.94K	32.34K	942	0.10
Canada	45.22K	12.85K	169	0.28

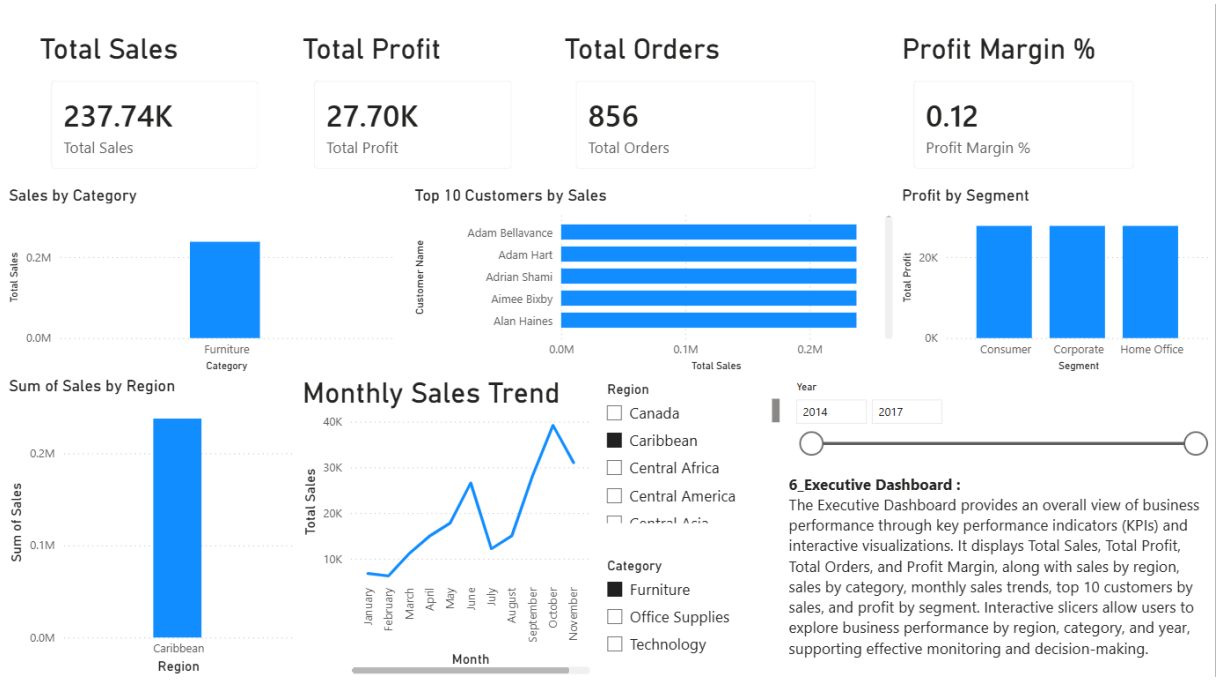


Figure 5: Executive Dashboard filtered to the Caribbean region.

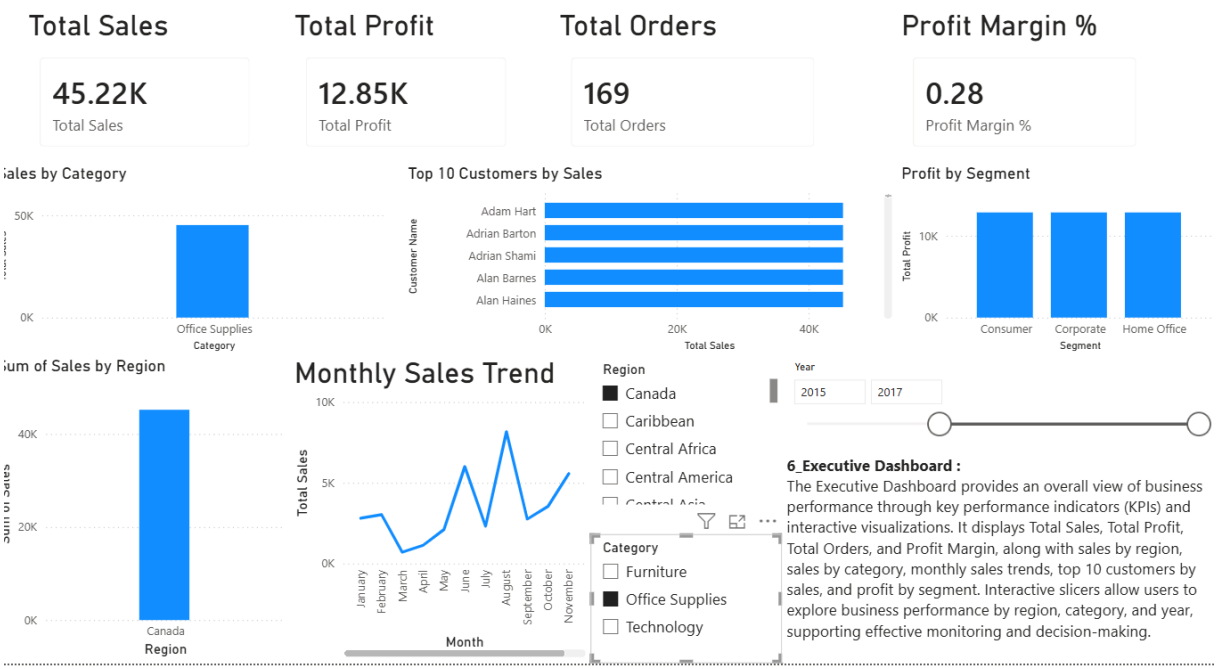


Figure 6: Executive Dashboard filtered to Canada, showing the highest margin (0.28) on low sales volume.

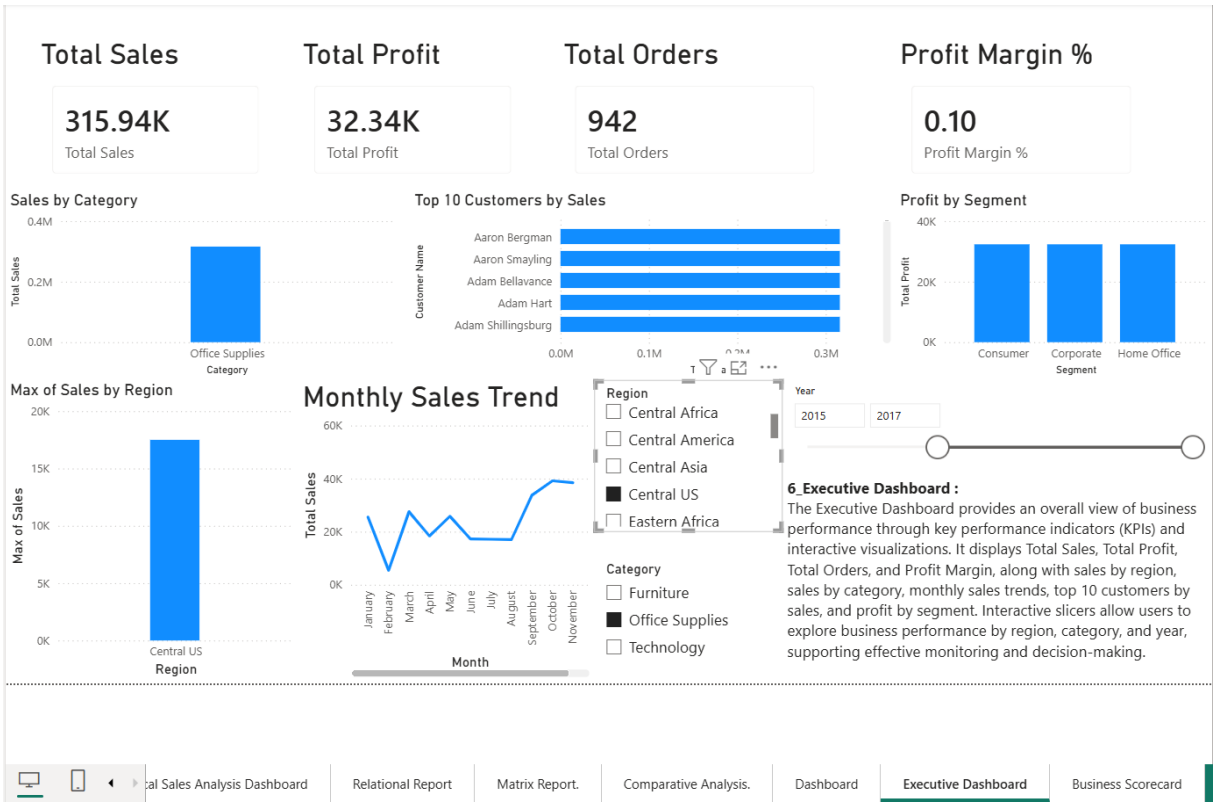


Figure 7: Executive Dashboard filtered to Central US, showing a below-average margin of 0.10.

Step 3: Analyzed Profitability by Product Category using Sales by Category and Profit by Category charts (Category on X-axis, Profit on Y-axis).

Observation (data model validation): In every report page the three categories, Furniture, Office Supplies and Technology, show the same value, and each bar equals the grand total (about 9.6M in Sales by Category; about 32K in Profit by Category when filtered to Central US). The same pattern appears in Sum of Sales by Segment, Profit by Segment, and Top 10 Customers by Sales, where every customer bar shows almost the same total. Selecting a Category in the slicer also leaves the KPI cards unchanged (Office Supplies with 2015 to 2017 still shows 7.91M, the total for all categories).

This shows that Category, Segment and Customer fields are not filtering the Sales and Profit measures through the current relationships, while the Region and Year fields filter correctly. In the Matrix Report, every country inside a region also repeats its region total (for example Afghanistan, Bangladesh and Bhutan each show 659,911.20, the Southern Asia total), and the Relational Report profit total (2,388,101.35) does not match the true total profit (1,101,816.89). The current model therefore relates the sales fact table to the other tables at Region level only.

Category-level and segment-level profitability conclusions are therefore **not drawn from the current visuals**, because doing so would report numbers that the data does not support. Correcting these relationships is recorded as a weakness and as a priority recommendation.

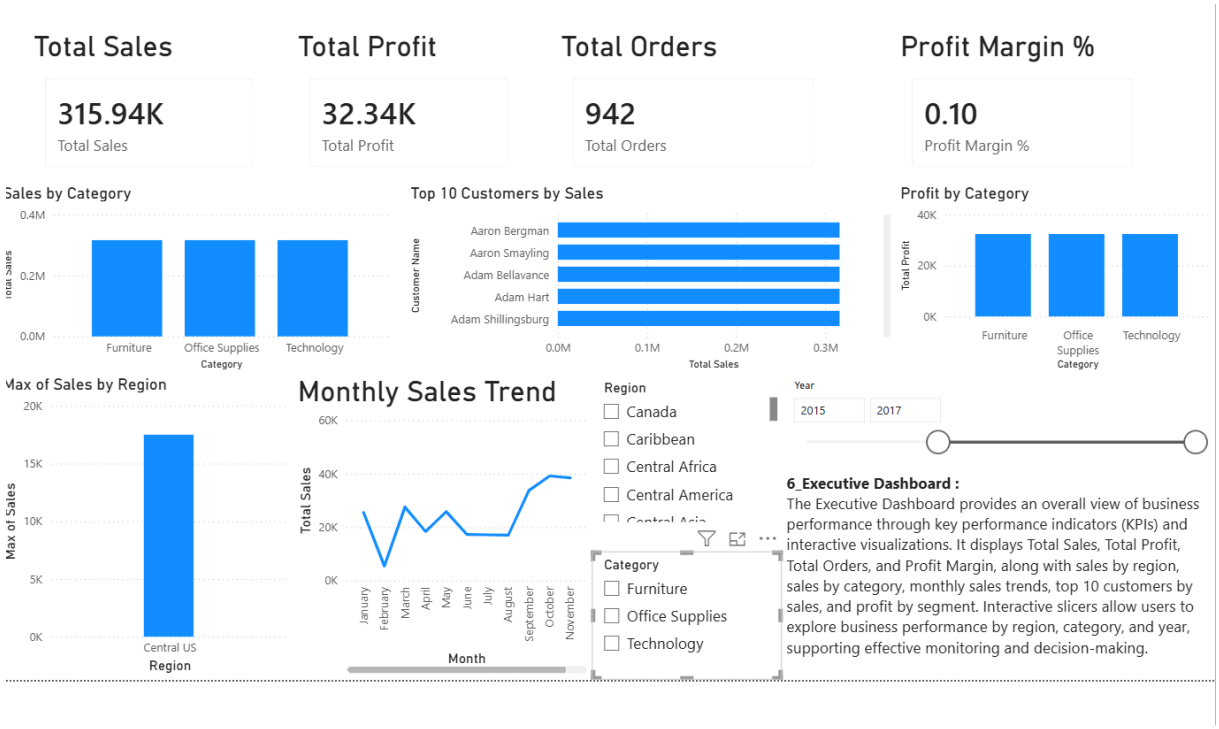


Figure 8: Sales by Category and Profit by Category show identical bars for all three categories, revealing a relationship issue in the data model.

Step 4: Analyzed Customer Segment Performance using Profit by Segment and Sum of Sales by Segment. Consumer, Corporate and Home Office show identical values (each bar equals the filtered total, for example about 917K profit for 2015 to 2017). For the reason explained in Step 3, segment comparison cannot be made reliably until the model relationships are corrected.

Step 5: Analyzed Monthly and Yearly Sales Trends using the Sales Trend by Year line chart and the Monthly Sales Trend chart.

Year	Total Sales (approx.)	YoY growth	Evidence
2014	1.69M	-	Total 9.60M minus 2015 to 2017 total of 7.91M
2015	2.03M	+20%	Sales Trend by Year chart
2016	2.60M	+28%	Sales Trend by Year chart
2017	3.27M	+26%	Sales Trend by Year chart

- **Sales are increasing every year**, rising from about 1.69M in 2014 to about 3.27M in 2017, an increase of roughly 94% over the period.
- **Profit is not growing faster than sales.** The Sales Grow, Margin Stays Flat chart on the Business Insights page shows the yearly margin at about 10.9% (2014), 11.4% (2015), 11.9% (2016) and 11.5% (2017). The margin stays within a narrow 11 to 12% band, and in 2017, the year with the highest sales, it actually fell. This confirms management's concern: the business is growing revenue, but it is not converting that growth into higher profitability.
- **Monthly seasonality (2015 to 2017):** February is the weakest month (about 0.35M) and sales build through the second half of the year, with peaks in June (about 0.82M), September (about 0.86M) and November (about 0.95M, the highest). July shows a clear dip (about 0.48M).

Note: monthly and yearly values are read from the line charts and are approximate.

Step 6: Documented significant observations and captured screenshots (Win + Shift + S) of the Executive Dashboard, regional analysis, category and segment visuals, sales trends, and Business Scorecard. These are included as Figures 1 to 10 in this report and saved in the Experiment_7_Screenshots folder.

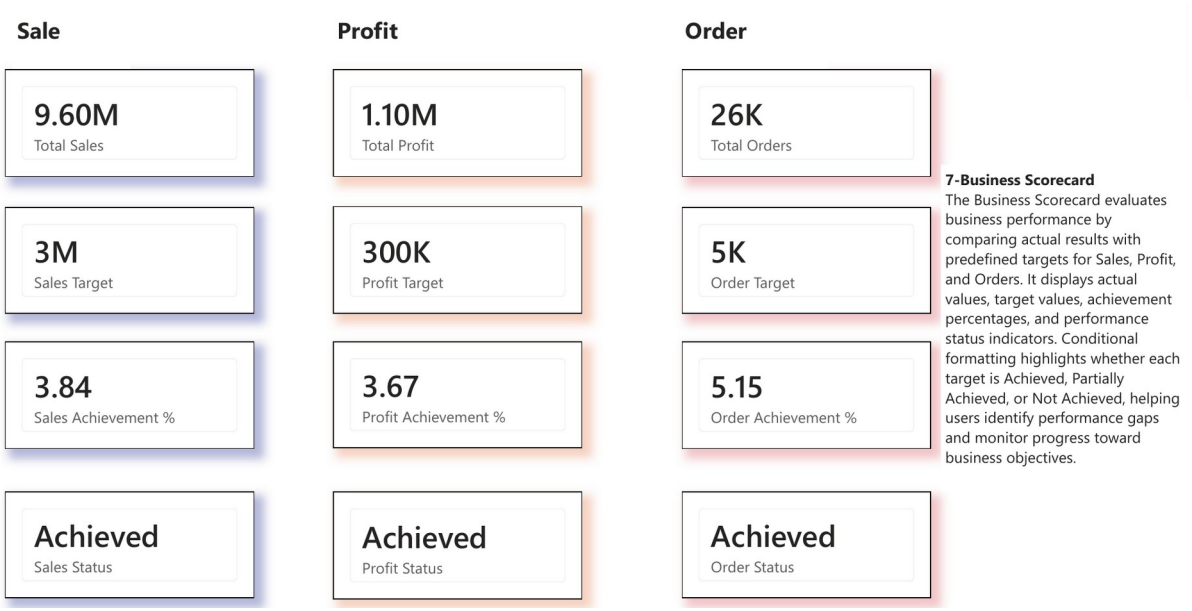


Figure 9: Business Scorecard: Sales 9.60M vs 2.5M target (3.84), Profit 1.10M vs 300K target (3.67), Orders 26K vs 5K target (5.15).

Scorecard observation: All three targets show "Achieved" with achievement of 384% (Sales), 367% (Profit) and 515% (Orders). The Sales Target card shows "3M" because of display rounding; the actual target is 2.5M (9.60M / 2.5M = 3.84). The targets are illustrative and far below actual performance, so the scorecard currently cannot separate strong and weak performance. Achievement values are also displayed as decimals (3.84) instead of percentages (384%).

Part B: Business Problem Identification

Step 7: Identified Business Strengths.

Strength	Evidence from Power BI	Business significance
S1. Consistent sales growth	Sales Trend by Year rises every year: about 1.69M (2014) to 3.27M (2017), YoY growth of 20%, 28% and 26%.	Demand is strong and growing; the business has a solid revenue base to build on.
S2. Western Europe is the leading region on both sales and profit	Sales 1,294,590.88 (13.5% of total) and Profit 150,495.79 (13.7% of total), margin 11.6%.	The largest market is also the largest profit source, so growth here directly adds profit.
S3. Strong contribution from Eastern Asia, Central America and Southern Asia	Eastern Asia: about 634K sales, 122.8K profit (19.4%). Central America: 934K sales, 120K profit (12.9%). Southern Asia: 660K sales, 117K profit (17.7%).	Together these three regions supply about 33% of company profit at above-average margins.
S4. Overall profitability is positive and stable	Profit 1.10M on 9.60M sales; margin 11.5%; Profit Achievement 367% of target.	The business is fundamentally profitable, so the problem is margin improvement, not survival.

Step 8: Identified Business Weaknesses.

Weakness	Evidence from Power BI	Possible implication
W1. Loss-making regions	Western Asia profit -39,604.61 (-16.6% margin); Western Africa -37,294.52 (-27.5%); Central Asia -46.4% margin on small sales. Western Asia and Western Africa alone lose about 76.9K.	These regions remove about 7% of total profit. Pricing, discounting, shipping cost or product mix in these markets needs investigation.
W2. High sales but very low margin	Southeastern Asia: 665K sales (6.9% share) but only 22.4K profit (2.0% share, 3.4% margin). South America: 459K sales, 18.0K profit (3.9%).	Revenue in these regions is not converting into profit, which is a direct cause of profit lagging behind sales.
W3. Flat profit margin despite rising sales	Sales nearly doubled from 2014 to 2017, but the yearly margin moved only between about 10.9% and 11.9%, and fell to about 11.5% in 2017.	Growth is volume-driven, not margin-driven. This is the core problem in the business scenario.
W4. Analytical gaps in the BI model and scorecard	Category, Segment and Customer visuals show identical values; scorecard targets are achieved by 367% to 515%.	Management cannot yet see which products or customer types lose money, and the scorecard does not flag real performance gaps.

Step 9: Identified Business Opportunities.

Opportunity	Supporting evidence	Proposed action	KPI to monitor
O1. Lift margin in Southeastern Asia and South America	Combined sales 1.12M at 3.4% to 3.9% margin vs 11.5% company average.	Review discounts, pricing and shipping costs. At the company-average margin these regions would earn about 88.7K more profit (54.0K + 34.8K).	Profit Margin %, Region Profit
O2. Turn around or restructure loss-making regions	Western Asia and Western Africa lose about 76.9K combined; Central Asia runs at -46.4%.	Investigate loss-making orders and products; cap discounts; stop or reprice unprofitable lines. Reaching break-even alone adds about 76.9K profit.	Region Profit
O3. Grow high-margin regions	Central Africa 25.5%, North Africa 24.8%, Eastern Europe 24.2%, Canada 28% margin, but each has under 2.5% of sales.	Targeted marketing and wider product range in these markets to grow volume at high margin.	Region Sales, Sales Share %
O4. Plan around seasonality	Peaks in June, September and November; lowest month February; dip in July.	Align inventory, staffing and promotions with peak months; run full-price campaigns in slow months instead of deep discounts.	Monthly Sales, Monthly Profit

Step 10: Supported findings with analytical evidence. The evidence used for each analysis is summarized below.

Analysis	Power BI visual used	Key evidence recorded
Regional performance	Sum of Sales by Region, Profit by Region, Matrix Report	Western Europe leads (1.29M / 150.5K); three loss-making regions
Category profitability	Sales by Category, Profit by Category	Identical bars: relationship issue identified
Customer performance	Profit by Segment, Top 10 Customers by Sales	Identical bars: relationship issue identified
Sales trends	Sales Trend by Year, Monthly Sales Trend	1.69M to 3.27M growth; flat 11 to 12% margin; Q4 peak
Overall performance	KPI cards on Executive Dashboard	9.60M sales, 1.10M profit, about 25.7K orders, 11.5% margin
Scorecard	Business Scorecard page	384% / 367% / 515% achievement; all Achieved
Margin analysis	Business Insights page	Yearly margin 10.9% to 11.9%; margin by region from -46.4% to 25.5%

Temporary slicer filters were cleared before capturing the overall-performance screenshots.

To bring this evidence together in one place, a new report page named **Business Insights** was added to the Power BI file. It contains:

- **Sales Grow, Margin Stays Flat:** a line and clustered column chart with Year on the X-axis, Total Sales as columns and Profit Margin % as a line.
- **Profit Margin % by Region:** a clustered bar chart sorted ascending, with rule-based conditional formatting: red for negative margins, orange for margins below 5%, and blue for the rest.
- **Sales vs Profit by Region:** a scatter chart with Region as values, Total Sales on the X-axis and Total Profit on the Y-axis, to expose high-sales, low-profit regions.
- **Region detail table:** Total Profit (with data bars), Profit Margin %, Region and Total Sales.
- **Key Insights text box** summarizing the findings and the main recommendation, together with Region and Year slicers.

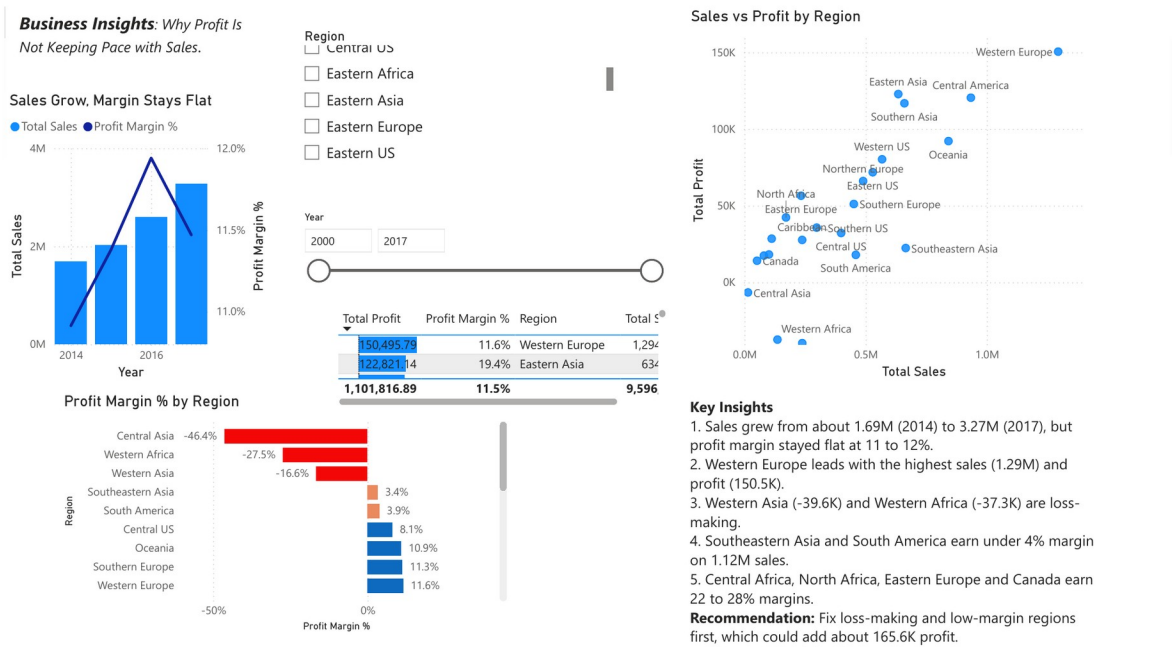


Figure 10: Business Insights page: yearly margin trend, Profit Margin % by Region, Sales vs Profit scatter, region table and Key Insights.

The page confirms the regional findings at a glance: Central Asia (-46.4%), Western Africa (-27.5%) and Western Asia (-16.6%) appear in red, Southeastern Asia (3.4%) and South America (3.9%) in orange, and Southeastern Asia sits far to the right but low on the scatter chart. Eastern Asia appears high on the scatter chart with 122,821.14 profit and a 19.4% margin.

Part C: Business Insight Report

Step 11: Prepared the Business Insight Report with the sections below.

C.1 Objective of Analysis

To analyze Global Superstore business performance using the existing Power BI reports and dashboards, explain why profitability is not growing at the same rate as sales, identify areas of strong and weak performance, and recommend data-driven improvements.

C.2 Reports and Dashboards Used

- **Geographical Sales Analysis Dashboard** (Experiment 5): sales by region, segment and category with geographic filters.
- **Matrix Report** (Experiment 5): Total Sales, Total Profit and Quantity by country and region.
- **Comparative Analysis** (Experiment 5): Profit by Region, Sales Trend by Year, Sales by Category, Top 10 Customers.
- **Executive Dashboard** (Experiment 6): KPI cards, five charts, Region/Category/Year slicers.
- **Business Scorecard** (Experiment 6): actual vs target, achievement % and status for Sales, Profit and Orders.
- **Business Insights** (Experiment 7, new page): yearly margin trend, margin by region, Sales vs Profit scatter, region table and key insights.

C.3 Key Findings

1. **Sales are growing strongly but margin is flat.** Sales rose from about 1.69M (2014) to 3.27M (2017), while profit margin stayed between about 10.9% and 11.9% and dipped in 2017. Profit grows only because volume grows. (Sales Grow, Margin Stays Flat chart)
2. **Western Europe is the strongest region.** It leads both sales (1.29M) and profit (150.5K). (Matrix Report, Profit by Region)
3. **Three regions lose money.** Western Asia (-39.6K) and Western Africa (-37.3K) remove about 7% of total profit, and Central Asia has the worst margin of any region (-46.4%). (Matrix Report, Business Insights page)
4. **Eastern Asia is an under-recognized strength.** It earns 122.8K profit at a 19.4% margin, second only to Western Europe in profit. (Business Insights page)
5. **Two large regions earn very little.** Southeastern Asia and South America generate 1.12M sales (11.7% of total) but only 40.4K profit (3.7% of total). (Matrix Report)
6. **Small markets are highly profitable.** Central Africa, North Africa, Eastern Europe and Canada earn 24% to 28% margins on small volumes. (Matrix Report, Canada filter)
7. **Sales are seasonal.** Second-half months, especially November, are strongest; February is weakest. (Monthly Sales Trend)
8. **The BI model needs correction.** Category, Segment and Customer visuals cannot currently separate performance, and scorecard targets are too low to flag gaps. (Profit by Category, Business Scorecard)

C.4 Business Strengths

Consistent year-on-year sales growth (20% to 28%), Western Europe as a leading high-sales and high-profit region, strong profit contribution from Eastern Asia, Central America and Southern Asia, and a stable positive overall margin of 11.5% (see Step 7).

C.5 Business Weaknesses

Loss-making operations in Western Asia, Western Africa and Central Asia, very low margins in Southeastern Asia and South America, a flat profit margin despite rapid sales growth, and gaps in the BI model and scorecard that hide product and customer-level problems (see Step 8).

C.6 Opportunities for Improvement

Improve margins in high-volume low-margin regions, restore loss-making regions to break-even, expand in high-margin small markets, and use seasonal patterns for planning and promotions (see Step 9). Bringing Southeastern Asia and South America to the company-average margin and loss-making regions to break-even would add an estimated 165.6K of profit, about 15% more than today's 1.10M, without any increase in sales.

C.7 Recommendations

Identified issue	Recommendation	Expected benefit	KPI to monitor
Loss-making Western Asia, Western Africa and Central Asia	Drill down to country, product and order level to find loss-making lines; set discount limits and reprice or remove unprofitable products.	Up to about 76.9K profit recovered at break-even	Region Profit, Profit Margin %
Low margin in Southeastern Asia and South America	Review discount policy, shipping costs and product mix; move sales toward higher-margin products.	About 88.7K extra profit at company-average margin	Region Profit Margin %
Flat margin despite sales growth	Set margin-based targets, not only sales targets; reward profitable growth rather than volume.	Profit grows faster than sales	Profit Margin % by Year
Untapped high-margin markets	Targeted marketing and product range expansion in Central Africa, North Africa, Eastern Europe and Canada.	Higher-margin sales growth	Region Sales, Sales Share %
Seasonal demand	Plan inventory and staffing for June, September and November peaks; use non-discount campaigns in February and July.	Smoother revenue and fewer margin-reducing discounts	Monthly Sales and Profit
Model and scorecard gaps	Fix relationships between the sales fact table and the Product/Customer tables using Product ID, Customer ID and Order ID keys; reset targets using prior-year actuals plus a growth rate; format achievement as %.	Reliable category, segment and customer analysis; meaningful scorecard status	Data validation checks, Achievement %

C.8 Conclusion

Global Superstore is a growing and profitable business, with sales nearly doubling between 2014 and 2017 and Western Europe leading both sales and profit. However, the profit margin has stayed at about 11 to 12%, which explains why profit is not growing as fast as management expects. The main causes visible in Power BI are three loss-making regions and two high-volume regions with margins below 4%. Fixing these, growing high-margin markets, and correcting the data model so that category and customer profitability can be analyzed are the priority actions. Power BI made it possible to locate these issues quickly and attach measurable KPIs to each recommendation.

Step 12: Inserted the Power BI screenshots (Figures 1 to 10) with descriptive captions to support each finding.

Part D: Executive Presentation

Step 13: Prepared a 6-slide Executive Presentation (BI_Experiment_7_Executive_Presentation.pptx) with the following structure.

Slide	Title	Content
1	Introduction	Experiment title, business scenario (sales up, profit not keeping pace), objective, Global Superstore dataset.
2	Dashboard Overview	Executive Dashboard screenshot (Figure 1); KPIs: 9.60M sales, 1.10M profit, about 25.7K orders, 11.5% margin.
3	Key Findings	Sales 1.69M to 3.27M with flat margin; Western Europe leads; Profit by Region chart (Figure 3).
4	Business Issues Identified	Strengths, weaknesses and opportunities summary; loss-making and low-margin regions with figures.
5	Recommendations	Top 3 actions with issue, action and KPI; estimated profit gain of about 165.6K.
6	Conclusion	Overall assessment, most important issue (flat margin), priorities, and the role of Power BI.

Step 14: Presented the findings before the instructor and class using the Executive Dashboard and Business Scorecard as live evidence.

Step 15: Justified each recommendation with data-driven evidence: every recommendation is linked to an exact figure from the Matrix Report, a trend from the line charts, or a KPI card, and to a KPI that will be used to measure the result.

Recommendation Summary Sheet

No.	Business finding	Supporting evidence	Recommendation	KPI to monitor
1	Loss-making regions	Western Asia -39.6K; Western Africa -37.3K; Central Asia -46.4% margin	Drill down, cap discounts, reprice or remove loss lines	Region Profit
2	High sales, low margin	SE Asia 3.4%; South America 3.9% margin	Review pricing, discounts, shipping and mix	Profit Margin %

No.	Business finding	Supporting evidence	Recommendation	KPI to monitor
3	Flat margin with rising sales	Yearly margin only 10.9% to 11.9%; dipped in 2017	Add margin targets to the scorecard	Profit Margin % by Year
4	High-margin small markets	Central Africa 25.5%; Canada 28%	Targeted growth campaigns	Region Sales
5	Seasonal demand	Peak November; low February and July	Seasonal inventory and promotion plan	Monthly Sales
6	BI model and scorecard gaps	Identical Category/Segment bars; 367% to 515% achievement	Fix relationships; recalibrate targets	Achievement %

Part E: Summary and Interpretation

Deliverable	Content	Purpose
Business Insight Report	Objective, reports used, findings, strengths, weaknesses, opportunities, recommendations, conclusion, screenshots	Explains business performance and the reasons behind it.
Executive Presentation	6 slides from introduction to conclusion	Communicates the findings to management concisely.
Recommendation Summary Sheet	Finding, evidence, recommendation, KPI	Gives management a one-page action list.
BI_Experiment_7.pbix	Existing reports, Executive Dashboard, Business Scorecard, new Business Insights page	Provides the supporting analytical evidence.

7. RESULT

Advanced analytics techniques were successfully integrated with the Business Intelligence reports and dashboards developed in Experiments 5 and 6 using Microsoft Power BI. KPI, trend, comparative, ranking and segmentation analysis showed that Global Superstore sales grew from about 1.69M in 2014 to about 3.27M in 2017, while the profit margin stayed flat at about 11 to 12%. A new Business Insights page was added to the Power BI file to present this evidence. Western Europe and Eastern Asia were identified as the strongest regions, Western Asia, Western Africa and Central Asia as loss-making regions, and Southeastern Asia and South America as high-sales, low-margin regions.

Business strengths, weaknesses, opportunities and recommendations were identified, supported by Power BI evidence, and communicated through a professional Business Insight Report, a Recommendation Summary Sheet, and a 6-slide Executive Presentation. The analysis also validated the data model and identified relationship and target-setting improvements needed for deeper category and customer analysis.

8. LEARNING OUTCOMES

- Apply advanced analytical techniques to business reports and dashboards. (CO3-BT3)

- Analyze organizational performance using BI tools and KPIs. (CO3-BT4)
- Generate business insights from analytical findings. (CO5-BT6)
- Develop strategic recommendations based on data-driven evidence. (CO5-BT6)
- Communicate analytical findings through professional reports and presentations. (CO5-BT6)
- Validate a BI data model by checking whether visuals respond correctly to filters and relationships.